ServiceNow ITSM Portfolio Project

IT Help Desk Analytics, Automation & Service Management

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Platform: ServiceNow Developer Instance (Zurich Release)

April 2026

Project Overview

This project demonstrates hands-on proficiency with the ServiceNow platform by building a complete IT Help Desk management system from scratch. Using a free ServiceNow Personal Developer Instance (PDI) on the Zurich release, three core ITSM components were designed and deployed:

- Real-Time Analytics Dashboard — 4-chart interactive dashboard monitoring open incidents across categories, priorities, states, and assignment groups
- Automated Workflow (Flow Designer) — Event-driven automation that sends an email alert whenever a Critical priority incident is created
- Service Catalog Item — A self-service employee request form for laptop provisioning with mandatory validation fields

The project simulates real-world enterprise IT operations and reflects skills directly applicable to ServiceNow Administrator, Business Analyst, and ITSM roles.

Part 1: Data Setup — Incident Generation

Before building any dashboards or reports, realistic incident data was required. Rather than manually creating records one by one, a JavaScript background script was written and executed directly in the ServiceNow server-side scripting console. The script automatically generated 50 mock IT incidents with randomized categories (Network, Hardware, Software, Database, Security), priorities (Critical, High, Moderate, Low), and states (New, In Progress, On Hold, Resolved, Closed).

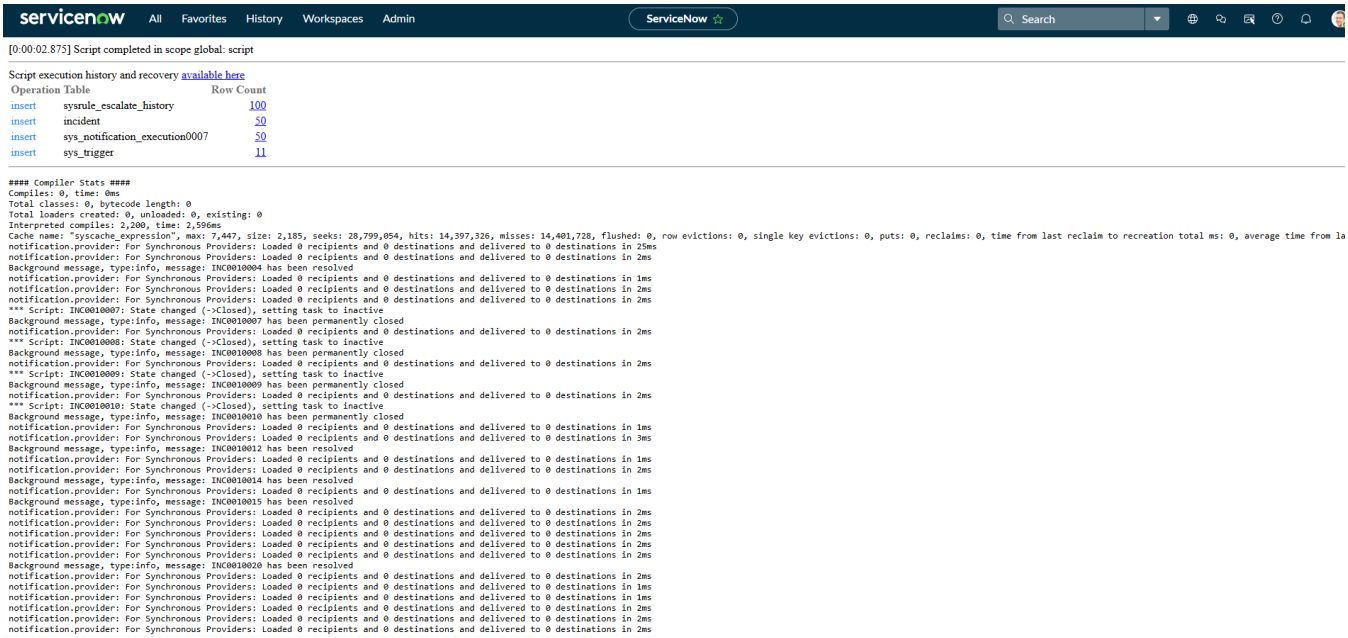

Figure 1: Background script execution confirming 50 incidents inserted into the Incident table

The script execution log confirmed successful insertion of 50 records. The resulting incident list is shown below, displaying real incident numbers, categories, priorities, and states — exactly as an IT manager would see them in a live production environment.

servicenow All Favorites History Workspaces Admin Incidents Search

Incidents Number Search Actions on selected rows... New

All

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010051	2026-04-04 12:48:53	Unable to connect to VPN	System Administrator	5 - Planning	On Hold	Hardware	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010050	2026-04-04 12:48:53	Email client crashing	System Administrator	5 - Planning	On Hold	Software	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010049	2026-04-04 12:48:53	Password reset required	System Administrator	5 - Planning	On Hold	Hardware	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010048	2026-04-04 12:48:53	Software license expired	System Administrator	5 - Planning	In Progress	Hardware	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010047	2026-04-04 12:48:53	Laptop screen not working	System Administrator	5 - Planning	In Progress	Database	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010046	2026-04-04 12:48:53	Unable to connect to VPN	System Administrator	5 - Planning	In Progress	Software	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010045	2026-04-04 12:48:53	Email client crashing	System Administrator	5 - Planning	Closed	Network	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010044	2026-04-04 12:48:53	Database connection timeout	System Administrator	5 - Planning	On Hold	Network	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010043	2026-04-04 12:48:53	Laptop screen not working	System Administrator	5 - Planning	New	Security	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010042	2026-04-04 12:48:53	Laptop screen not working	System Administrator	5 - Planning	Resolved	Hardware	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010041	2026-04-04 12:48:53	Slow internet connection	System Administrator	5 - Planning	Closed	Software	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010040	2026-04-04 12:48:53	Software license expired	System Administrator	5 - Planning	On Hold	Database	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010039	2026-04-04 12:48:53	Unauthorized access attempt detected	System Administrator	5 - Planning	Closed	Hardware	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010038	2026-04-04 12:48:53	Unable to connect to VPN	System Administrator	5 - Planning	Closed	Security	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010037	2026-04-04 12:48:53	Laptop screen not working	System Administrator	5 - Planning	Resolved	Security	(empty)	(empty)	2026-04-04 12:48:53	admin
INC0010036	2026-04-04 12:48:52	Laptop screen not working	System Administrator	5 - Planning	On Hold	Security	(empty)	(empty)	2026-04-04 12:48:52	admin
INC0010035	2026-04-04 12:48:52	Database connection timeout	System Administrator	5 - Planning	Closed	Software	(empty)	(empty)	2026-04-04 12:48:52	admin
INC0010034	2026-04-04 12:48:52	Laptop screen not working	System Administrator	5 - Planning	Closed	Network	(empty)	(empty)	2026-04-04 12:48:52	admin
INC0010033	2026-04-04 12:48:52	Slow internet connection	System Administrator	5 - Planning	New	Hardware	(empty)	(empty)	2026-04-04 12:48:52	admin
INC0010032	2026-04-04 12:48:52	Laptop screen not working	System Administrator	5 - Planning	Closed	Database	(empty)	(empty)	2026-04-04 12:48:52	admin

Figure 2: 50 mock IT incidents loaded into the ServiceNow Incident table

Part 2: Report Builder — Individual Charts

Four individual reports were created using the ServiceNow Report Builder. Each report was configured with a specific data source, visualization type, and grouping field to answer a distinct business question about incident data.

2.1 Report Configuration

Each report was built using the 4-step wizard: Data → Type → Configure → Style. The data source, table, applied conditions, and grouping field were configured at this stage.

Data > Type > Configure > Style

* Report name

Incidents by Category

* Source type

Data source

* Data source

Incidents.Closed (Incident)

Description

There is no description for this data source. To add a description, please contact your admin.

Table

Incident

Applied conditions

Active = false

2.2 Incidents by Category (Pie Chart)

This pie chart breaks down closed incidents by IT category, revealing which service areas generate the most support tickets. Inquiry/Help dominates at 39.47%, followed by Software (23.68%) and Hardware (15.79%).

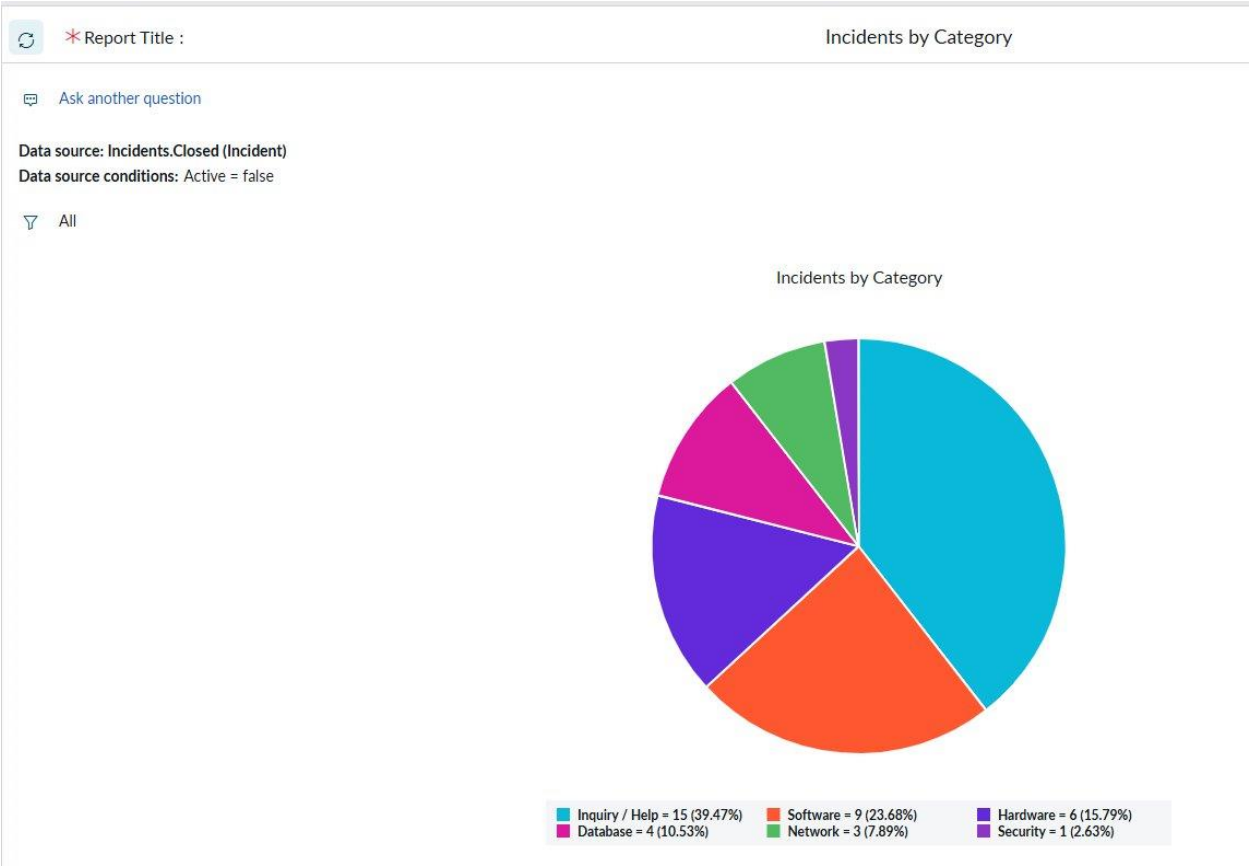

Figure 4: Incidents by Category — Pie Chart showing distribution across IT service categories

2.3 Incidents by Priority (Bar Chart)

This bar chart shows open incidents grouped by priority level, enabling IT managers to quickly identify backlogs of critical or high-priority work that requires immediate attention.

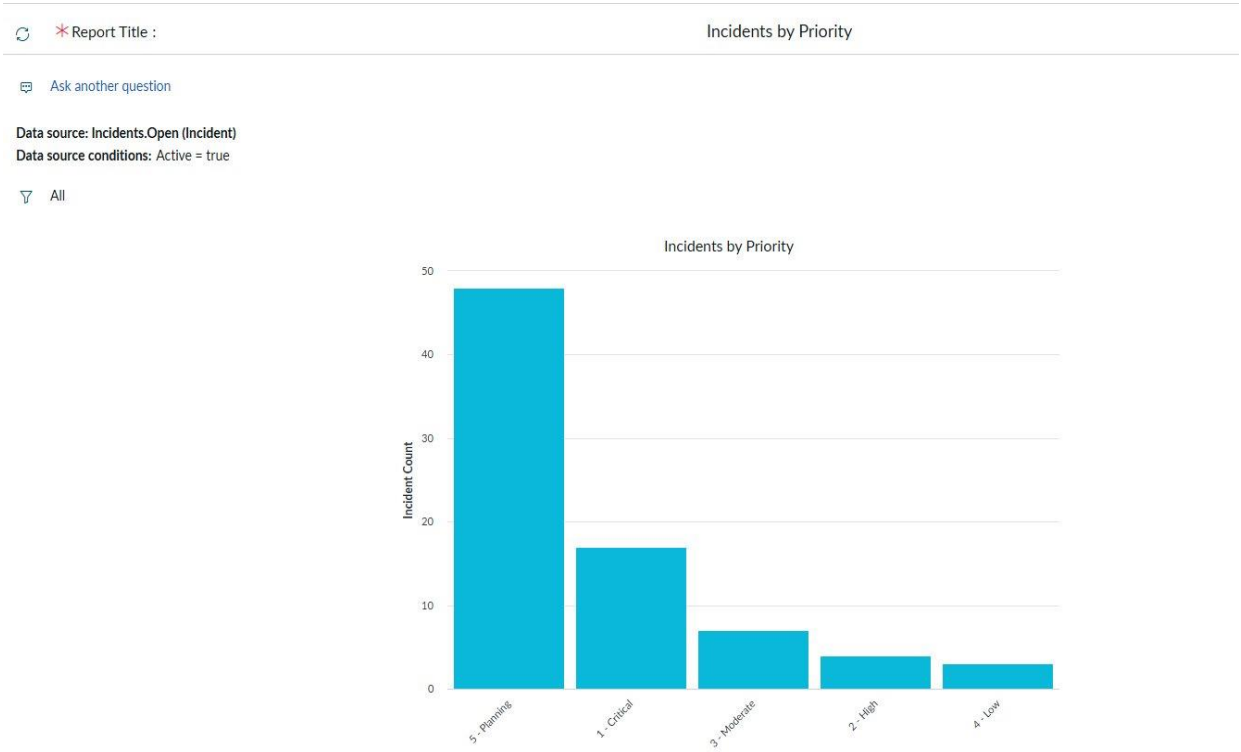

Figure 5: Incidents by Priority — Bar Chart of open incidents by priority level

2.4 Incidents by State (Donut Chart)

The donut chart visualizes the current state distribution of all open incidents — In Progress (36.71%), On Hold (27.85%), New (24.05%), and Resolved (11.39%). This gives instant visibility into workflow bottlenecks.

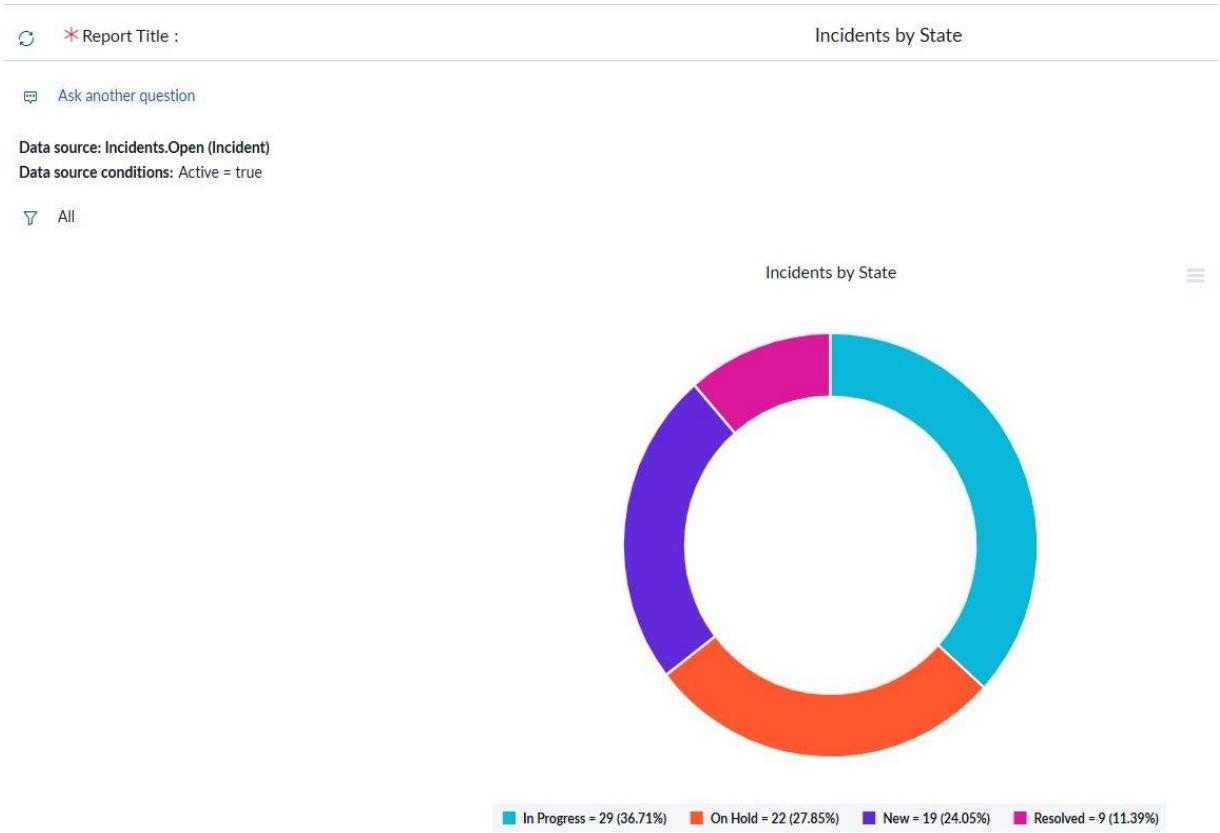

Figure 6: Incidents by State — Donut Chart showing incident lifecycle distribution

2.5 Incidents by Category (Bar Chart)

A second category-based chart using open incidents provides a bar view for easier magnitude comparison between categories, complementing the pie chart with a different visual perspective.

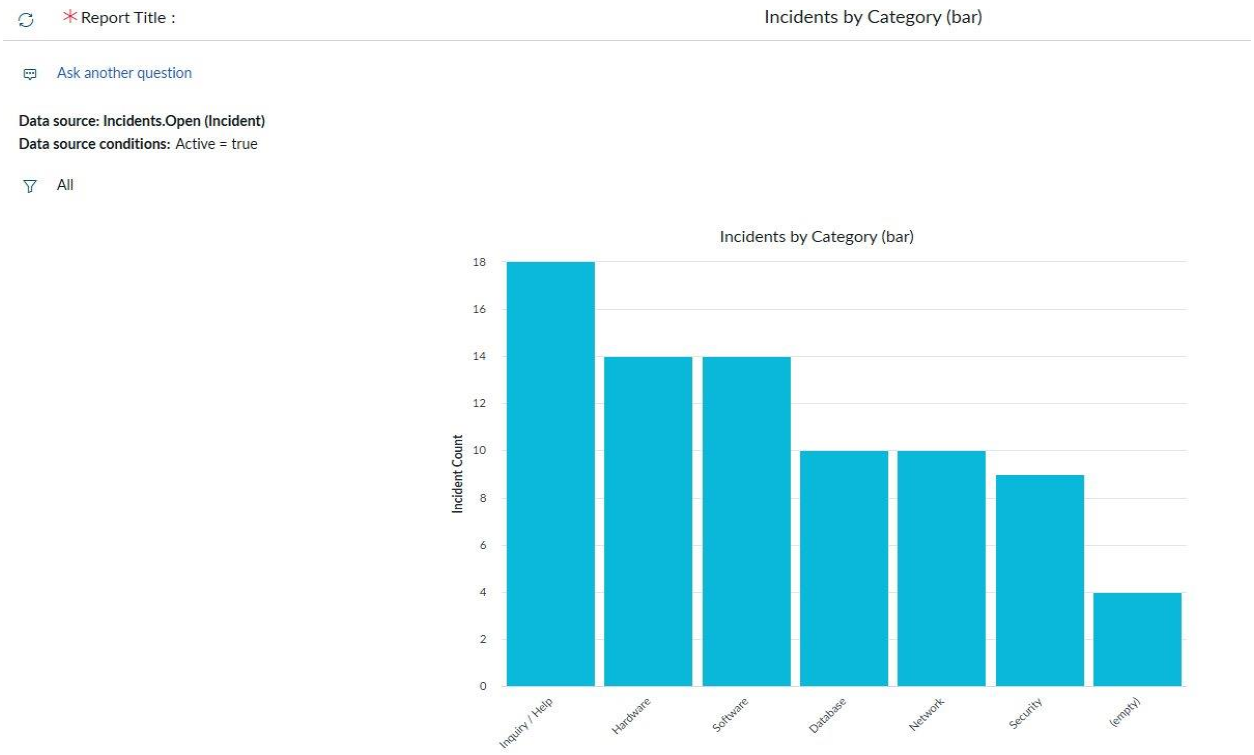

Figure 7: Incidents by Category (Bar) — Open incident counts by IT service category

Part 3: IT Help Desk Analytics Dashboard

All four charts were assembled into a single Platform Analytics dashboard titled "IT Help Desk Analytics Dashboard | Real-Time Incident Monitoring." A Category filter (slicer) was added to allow dynamic filtering of all visualizations simultaneously — similar to slicers in Power BI or Tableau.

The dashboard provides a complete at-a-glance view of the IT help desk's operational status, enabling data-driven decisions about resource allocation, escalation priorities, and workload distribution.

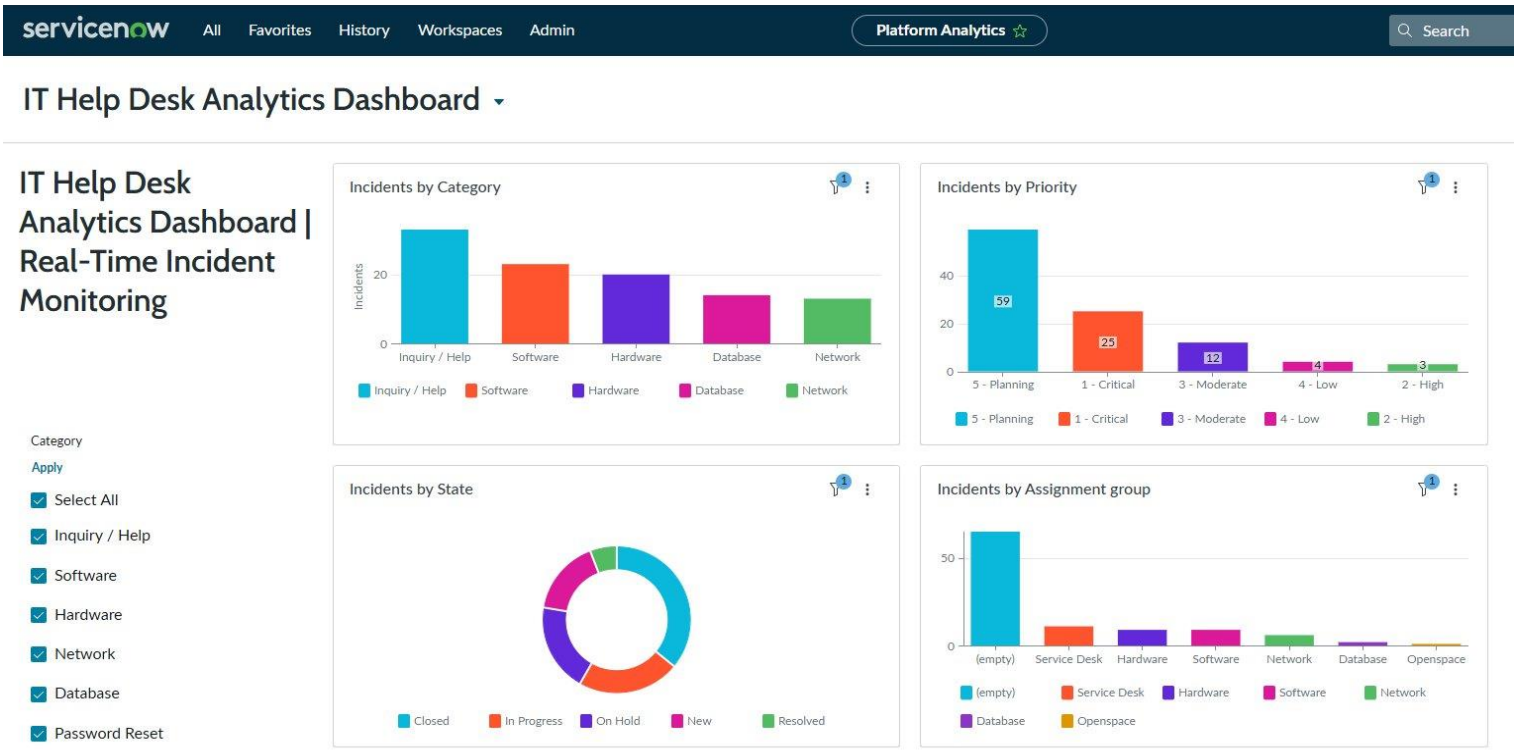

Figure 8: Final IT Help Desk Analytics Dashboard with 4 charts and Category filter

Part 4: Automated Workflow — Critical Incident Alert

An automated workflow was built using ServiceNow Flow Designer to demonstrate event-driven automation. The flow triggers automatically when a new incident is created and the priority is set to 1 - Critical. When this condition is met, the flow immediately sends an email alert to the IT manager.

This eliminates the need for manual monitoring of the incident queue and ensures that Critical incidents receive immediate attention — a key capability in enterprise ITSM environments.

- Trigger: Incident Created (where Priority = 1 - Critical)
- Action: Send Email to IT Manager
- Status: Active and deployed on the developer instance

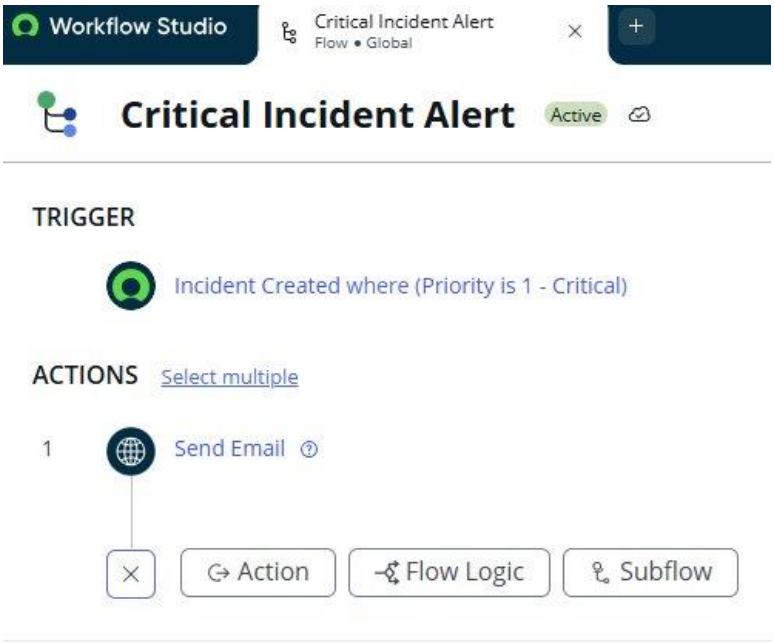

Figure 9: Flow Designer — Critical Incident Alert workflow (Active)

Part 5: Service Catalog — New Employee Laptop Request

A Service Catalog item was created to demonstrate self-service request management — a core component of enterprise ITSM. The "New Employee Laptop Request" item allows employees to submit a standardized laptop request through the ServiceNow portal, replacing ad-hoc email requests with a structured, trackable workflow.

5.1 Catalog Item Configuration

The item was configured under the Hardware category in the Service Catalog with a 2-day delivery SLA, full description, and standard pricing.

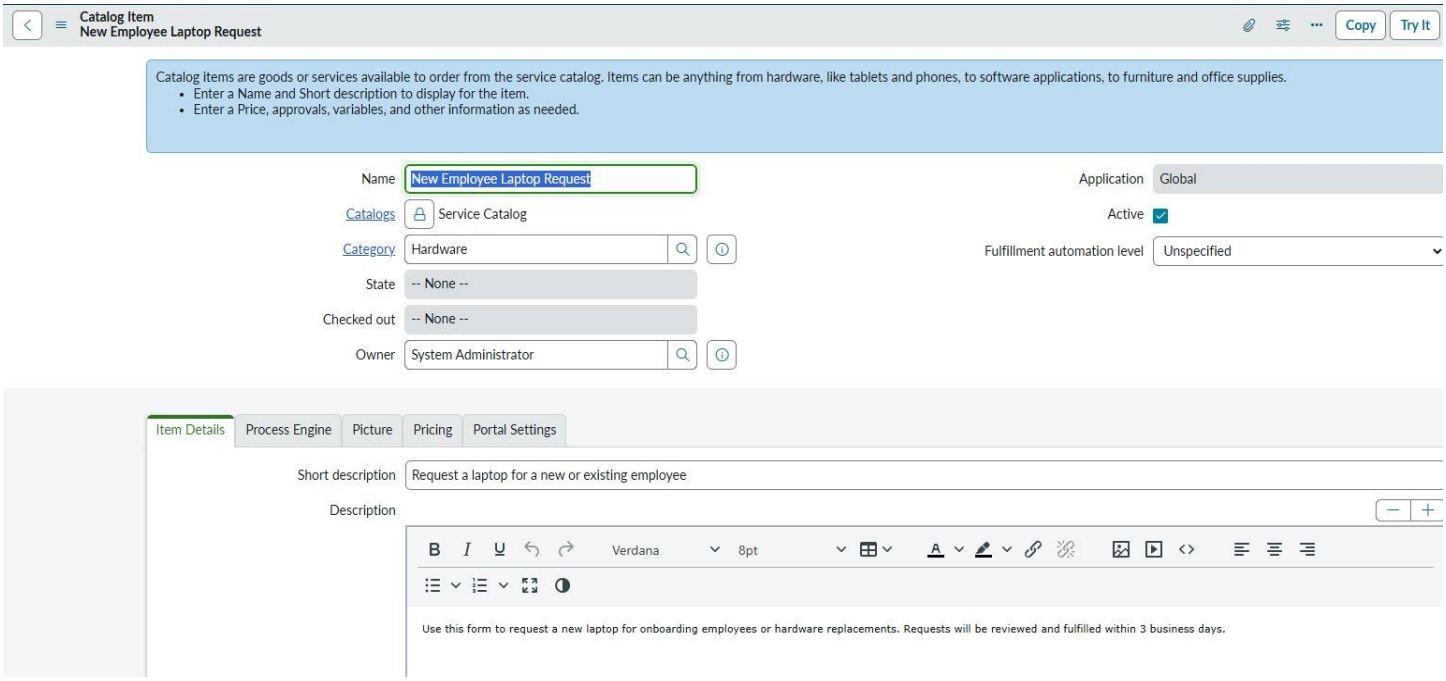

Figure 10: Catalog Item configuration — New Employee Laptop Request

5.2 Form Variables

Three mandatory form variables were added to capture the information needed to fulfill the request: employee full name, required laptop type, and needed-by date.

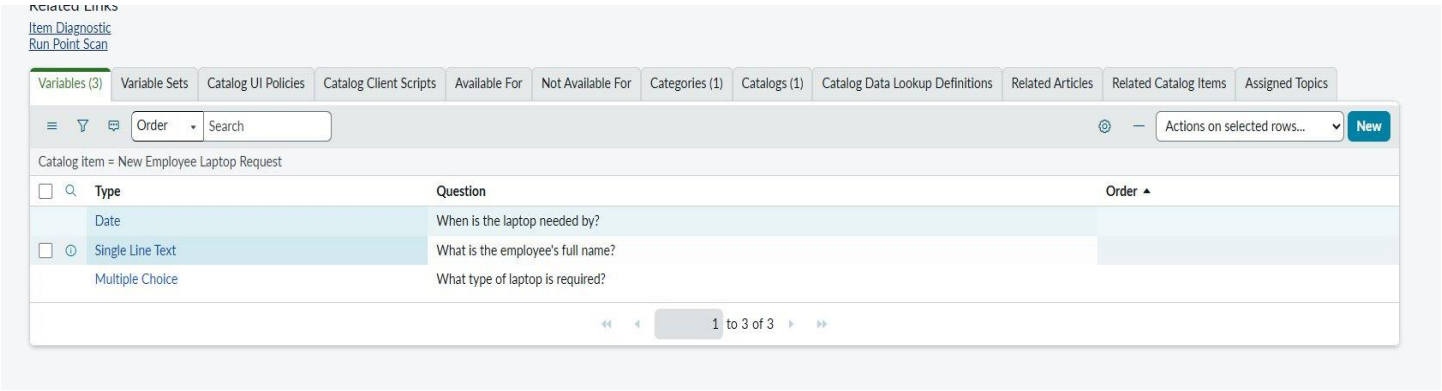

Figure 11: Catalog Item variables — 3 form fields configured for the request form

5.3 Live Employee-Facing Form

The final form as seen by an employee in the Service Catalog portal. The form includes a date picker, text input for the employee name, and a multiple choice selector for laptop type — with an Order Now button and shopping cart functionality.

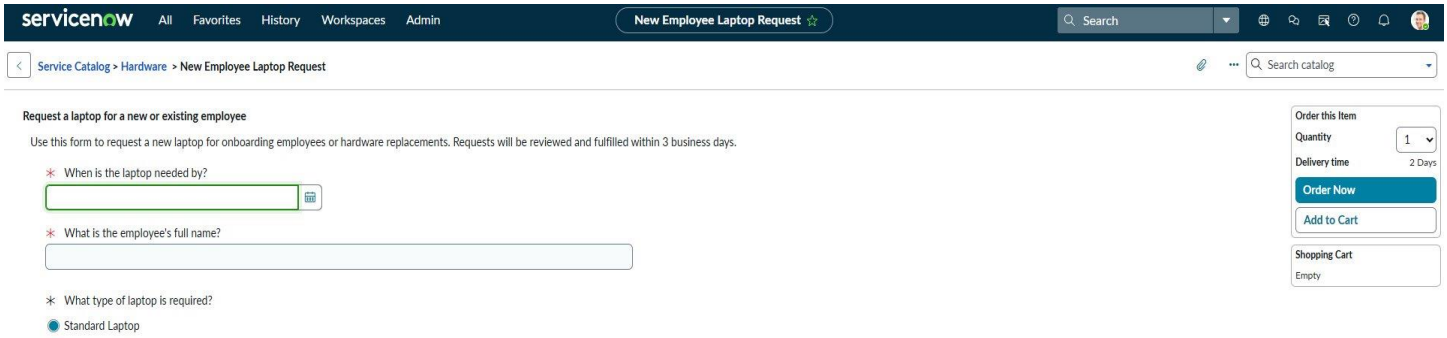

Figure 12: Live Service Catalog form — New Employee Laptop Request as seen by employees

Skills Demonstrated

ServiceNow Feature	What Was Built
Background Scripting	JavaScript server-side script to generate 50 realistic mock IT incidents
Incident Management	Configured incidents with categories, priorities, states, and assignment groups
Report Builder	4 reports built: Pie, Bar (x2), and Donut charts with custom data sources
Platform Analytics	Interactive dashboard with 4 visualizations and a Category filter/slicer
Flow Designer	Automated email alert triggered on Critical incident creation
Service Catalog	Self-service laptop request form with 3 mandatory variables and 2-day SLA

This project was completed entirely from scratch on a free ServiceNow Personal Developer Instance, without prior ServiceNow experience, combining a Business Analytics background with platform administration and workflow automation skills.